

Simran Mann

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EDUCATION

Simon Fraser University

Sept 2022 – Dec 2026

Bachelor of Applied Science in Computing Science

GPA: 3.99/4.33

- **Relevant Courses:** Distributed Systems, Compilers, Computer Architecture, Operating Systems

TECHNICAL SKILLS

Programming Languages: C/C++ (multithreading, atomics, STL, RAII), Python, x86-64 Assembly, Bash

Systems & Concurrency: POSIX threads, std::thread, mutexes, atomics, MPI, lock-free progress guarantees

Tools & Platforms: Linux/Unix, Git, Docker, LLVM, STM32

EXPERIENCE

SFU Robot Soccer

May 2025 – Present

Firmware Developer

- Designed and implemented the basestation firmware, enabling robots to receive and parse commands through UART from the software team and relay commands via LoRa to the motherboards for each robot
- Contribute to motherboard firmware development and testing, ensuring real-time task scheduling and responses
- Collaborate in a multidisciplinary team spanning firmware, software, and electronics to achieve robust match-ready performance for the robots

NETGEAR

Sep 2024 – April 2025

Orbi Software Test Engineer

- Designed and deployed multi-node network testbeds using managed switches, configuring VLANs, static routing, and network topologies to simulate mesh network environments
- Debugged network-level issues including packet loss, routing failures, and inter-device communication problems using Wireshark, tcpdump, ping, traceroute, and iperf for bandwidth analysis
- Validated router and mesh network behavior over TCP/UDP traffic, testing packet forwarding, protocol-level correctness, latency, and failover scenarios
- Developed Python automation tool to parse UART serial output, configure devices remotely, and automate network stability testing, reducing test execution time from 8 hours to 2 hours

PROJECTS

Parallel and Distributed Graph Algorithms

Sept – Dec 2024

- Implemented parallel PageRank and Triangle Counting algorithms using std::thread for shared-memory systems and MPI for distributed multi-node clusters
- Scaled implementations from single-node multithreaded execution to distributed MPI-based execution, optimizing message-passing to reduce synchronization overhead
- Evaluated vertex-based, edge-based, and dynamic task decomposition approaches to improve load balance and inter-process communication

Concurrent Data Structures

Oct – Nov 2024

- Implemented non-blocking queues, stacks, and sets using lock-free linked lists
- Benchmarked data structures under multi-producer/multi-consumer workloads to analyze contention bottlenecks
- Addressed ABA problems via version-tagged pointers and compare-and-swap (CAS) atomic operations

C Language Compiler

June – Aug 2024

- Developed compiler for C-like language using Flex/Bison and LLVM IR, including SSA-form code generation and short-circuit evaluation for booleans
- Implemented parsing, semantic analysis, and code generation pipelines
- Built C runtime and Python scripts to automate testing, validating compilation and execution of code

Custom Memory Allocator

Jul 2024

- Built malloc/free implementation from scratch with Best Fit, Worst Fit, and First Fit allocation algorithms
- Implemented memory compaction to reduce fragmentation and improve allocation performance
- Managed free and allocated memory blocks using custom linked list data structures